Technical review

Item #1 · captura.png · Run ID: 20261029T110057Z-95dba9

Path	`/tmp/tmp.sVTkOe7IbL/evidence/captura.png`
Size (bytes)	354
Last modified (UTC)	2026-10-29T11:00:57Z
Extension	.png
Detected type (signature)	PNG image
Status	No issues found
Case	SAMPLE-1
Evidence ID	E0001
Case registry	Verified: identical to the registered evidence
Registered at intake (UTC)	2026-10-29T11:00:57.109Z

Hashes

sha256	`666ed73f0863f60ec48123834935c56c7ec98487834fa366fc0d7be8a1fa03c5`

Findings

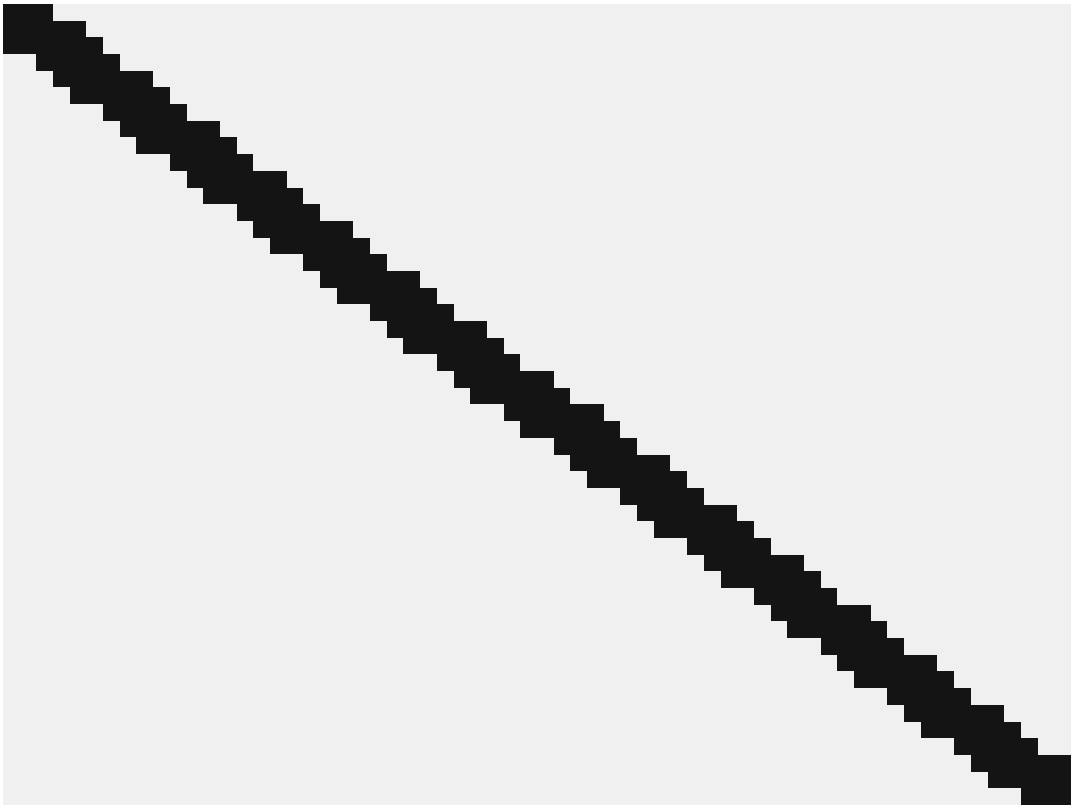
None

Image analysis

Status	Completed
Format	PNG
Dimensions (pixels)	64 × 48

Original image (decoded preview)

Lossless PNG of the decoded pixels, exactly as the analyses saw them (stored orientation; EXIF rotation not applied). The maps below have the same size and can be laid over it.



maps/original.png (64x48) — reduced copy for print; the full-resolution file is in the item folder

Metadata summary

Key fields only; metadata.json in the item folder holds every tag read. Metadata is written by the device or program that saved the file and can be edited or forged.

Tags read	0
Tags per group	—
GPS tags	absent
MakerNote	absent
XMP packet	absent
Embedded thumbnail	absent
ICC profile (SHA-256)	—

No metadata field was found.

Structure summary

Top-level segments in file order (JPEG markers, PNG chunks, RIFF chunks or HEIF boxes); structure.json holds the full tree with byte offsets.

Top-level segments	4
bit depth	8
colour type	2
bytes after IEND	0

Marker / chunk / box sequence:

```
signature · IHDR · IDAT · IEND
```

Fingerprint databases

Role	Database	Version	Entries	Status	SHA-256
starter	rakvor-starter	0	0	loaded	`e0b0c87e4b79e5c5402e920d343436056c25bb959a5bb757a7d0519b95ab4be8`
local	—	—	—	absent	—

Analysis maps

No analysis map was written for this item.

Copy-move shift vectors

Copy-move detection was not run (--copy-move off, or the pixels were not decoded).

Extracted files

structure.json (structure, 898 B)	sha256: c9f140622551085fda2b3428e0c6e81e54c4c6b13cdad57314638f4e6505974e
metadata.json (metadata, 123 B)	sha256: f07be8fa2c6e0057dc23806739e3091b649abb03e6cb9fe24458075adcbee13c
maps/original.png (original_preview, 908 B)	sha256: 348f1c293ad961931aea9209ad5587c18c4b4adf236eb33ba8b0058a9d158d94

Method

Each file was opened read-only and hashed in a single pass with the selected algorithms, and its real type was identified from its signature. JPEG, HEIC, PNG and WebP images were then re-read in an isolated process and the bytes read were hashed again and compared with the first pass. The image analysis read the file structure and metadata, compared the JPEG quantization tables with the IJG standard and the fingerprint databases, looked for double JPEG compression, wrote the selected visual maps (error level, noise, JPEG ghosts), searched for copied-and-moved regions, compared the embedded thumbnail with the main image and, when reference images were given, compared the image with them. HEIC pixels are not decoded (structure and metadata only). Every action and every file written was recorded in a hash-chained audit log.

Audit trail

Audit entries	#5 – #12
Log folder	`/tmp/tmp.sVTkOe7IbL/case/runs/rv-imgauth/20261029T110057Z-95dba9/logs`
Digital signature	This PDF is not digitally signed.

Limitations

- An image without findings is not proven authentic: careful edits can leave none of the traces these checks look for, and it is impossible to prove a negative (SWGDE 18-I-001).
- Error level, noise and ghost maps are visual aids for a trained examiner; the tool gives them no score and draws no conclusion from them.
- Double JPEG compression shows the image was saved more than once; ordinary sharing and exporting do this.
- The compression-table database covers few devices and apps (built from Rakvor's own captures plus local learning); an absent or matching entry is weak evidence on its own.
- Copy-move detection compares blocks at the same scale and orientation on a downscaled image: scaled, rotated or small copies are not found.

- Error rates of these checks have not yet been measured on a documented validation corpus; the findings are signals to be weighed by the examiner.
- Modification times come from the file system of the examined copy and may differ from the original source.
- A trusted timestamp proves the audit log existed in this exact form at the stated time; it does not prove that the examined copy matches the original source.
- Type detection uses a limited table of common file signatures; unknown types are reported as unknown, not as suspicious.